

OpenRocket Development: Triangular Fin Leading Edges

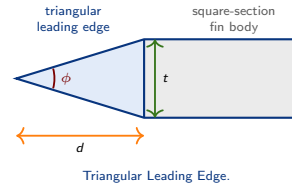
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Changes

- Added `FinSet.CrossSection.TRIANGULAR`: square-plate fin with a symmetric leading-edge bevel.
- Stored parameter is included angle ϕ ; UI can also edit vertex distance $d = (t/2) / \tan(\phi/2)$.
- Trapezoid, elliptical, and freeform fin dialogs get dynamic angle/distance controls and cross-section preview.
- .ork save/load persists triangular plus optional leading-edge angle in degrees.

Aerodynamic implementation

- Subsonic pressure drag: rounded-edge correlation at leading-edge-normal Mach.
- Supersonic pressure drag: attached symmetric wedge shock model; square stagnation fallback.
- Roll forcing/damping include leading-edge projected area; normal-force planform unchanged.
- RockSim/RASAero exports degrade to square; RASAero emits a warning.



Zephyrus comparison metric

Tool	Motor	Apogee	Time to apogee
RAS Aero	N8406-P	11620 ft	24.24 s
OpenRocket	N8406-P	11758 ft	25.70 s
OR - RAS Aero	same	+138 ft	+1.46 s

Verification added

- Tests cover pressure drag, base drag, angle-distance conversion, roll damping area, and .ork round trip.
- Compile/test checks run for core and Swing paths.